

# Neural Control of Bimanual Robots With Guaranteed Global Stability and Motion Precision

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## I. INTRODUCTION

**Abstract**—Robots with coordinated dual arms are able to perform more complicated tasks that a single manipulator could hardly achieve. However, more rigorous motion precision is required to guarantee effective cooperation between the dual arms, especially when they grasp a common object. In this case, the internal forces applied on the object must also be considered in addition to the external forces. Therefore, a prescribed tracking performance at both transient and steady states is first specified, and then, a controller is synthesized to rigorously guarantee the specified motion performance. In the presence of unknown dynamics of both the robot arms and the manipulated object, the neural network approximation technique is employed to compensate for uncertainties. In order to extend the semiglobal stability achieved by conventional neural control to global stability, a switching mechanism is integrated into the control design. Effectiveness of the proposed control design has been shown through experiments carried out on the Baxter Robot.

**Index Terms**—Bimanual robots, global uniformly ultimately boundedness (GUUB), neural networks (NNs), tailored tracking performance.

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WITH bimanual cooperation, humans are able to perform delicate and complicated manipulations. There has been a pronounced tendency in the robotics and automation community to shift focus of studies from single manipulators to coordinated dual-arm robots [1]–[6]. In comparison to a single-arm robot, a dual-arm robot has prominent advantages in handling capability, loading capability, as well as manipulative skills. For example, in tool using tasks such as carving or screwing, distribution of motions and forces required by the tasks between the two robot arms greatly reduces the complexity and energy cost of manipulation, compared with that of a single robot arm. Therefore, the topics of dual-arm robot control have attracted much research attention over the past decades [7]–[10]. The early studies of coordinative control schemes of two robotic arms were reported in [11] and [12], where the position tracking and force control were addressed. To deal with the unknown output hysteresis in the control of coordinate robot, an adaptive neural control was presented with computational efficiency [6]. In [7], a dual neural network (NN) has been used to resolve the distribution problem of redundant coordination robot systems by using multicriteria to minimize the global kinetic energy.

It should be emphasized that the motion precision is of great importance in the robot operation, especially for the dual-arm manipulation [13]. A precise coordination of both arms can ensure that no excessive internal force would occur, and also reduce possible variation of the internal forces. In this regard, the rigorous requirement of motion precision implies that the transient performance in the operation must also be taken into account. Therefore, much effort in the control community has been made to achieve a desired transient performance [14]–[17]. For this purpose, an effective tracking algorithm was proposed to control a five-bar closed-chain robot based on transformation of tracking errors in [16]. In [17], a constraint on output was considered for control of a class of multiinput multioutput (MIMO) systems. The above-mentioned control approaches rely on purposely built transformations with appropriate inverses, which increase the complexity of the control design.

In practice, usually, the kinematics information of robots can be accurately known from the manufacturer, but there exist inevitable uncertainties of the dynamics of the robot [18]–[22]. Nevertheless, we can always access the input–output data of a

robot system; thus, it is desirable to use available input–output data to approximate the unknown robot dynamics, in order to design a controller with satisfactory performance. One of the most successful control approaches is the NN based or fuzzy logics based intelligent controller, which utilizes the powerful universal approximation ability of them to compensate for unknown dynamics [23]–[34]. In [35], the NN was used to approximate the hypersonic flight vehicle dynamics in the tracking control of strict-feedback systems. In [15], the NN was used to compensate for the complicated nonlinearity in the closed-loop robot dynamics.

It should be noted that the above-mentioned NN control methods only ensure stability in the sense of semiglobally uniformly ultimately boundedness (SGUUB) of the closed-loop signals, because the NN's approximation only holds over a certain compact set, the so-called NN's approximation domain. Therefore, the range of state variable must be within this approximation domain during operation. However, such a compact set is impossible to be identified precisely beforehand, especially for highly nonlinear complicated systems with MIMO. Therefore, it is important to develop an NN controller with guaranteed global stability. In [36], a robust adaptive neural controller was developed to achieve global uniformly ultimately boundedness (GUUB) stability. An adaptive NN control for hypersonic flight vehicle systems was proposed to ensure GUUB stability in [35]. However, only single-input single-output systems were reported in most existing works, and few of them consider transient performance at the same time.

In this paper, we aim to achieve both tailored transient performance and guaranteed global stability at the same time, by exploiting the barrier Lyapunov functions (BLFs). The BLFs were originally developed in the nonlinear control community to deal with the state and output constraints [37]–[40]. A BLF-based controller was developed to control a robot manipulator with joint space constraints in [37]. In [40], an asymmetric time-varying BLF was presented for nonlinear systems in a strict-feedback form. In our previous work [41], a time-varying BLF based robot controller was developed to achieve desired tracking performance on a single arm robot in the joint space.

It is noted that by posing constraints to the behavior of the states or outputs, tracking errors can be indirectly constrained using the BLF technique. Motivated by this, in this paper, the BLF technique was exploited to achieve the tailored tracking performance at both transient and steady states. Compared with the regulation of steady-state responses, the shaping of the transient control is much more difficult. By constructing a prescribed tracking performance requirement function, a proper BLF is proposed for controller synthesis of a dual-arm robot, such that both transient and steady-state tracking performance can be ensured. Meanwhile, a switching mechanism is introduced into the NN controller design to ensure global stability. In comparison to the conventional NN controllers that only ensure the stability of SGUUB, our proposed NN controller guarantees global stability of the closed-loop system. This is practically much more useful as the requirement of the NN inputs is greatly relaxed.

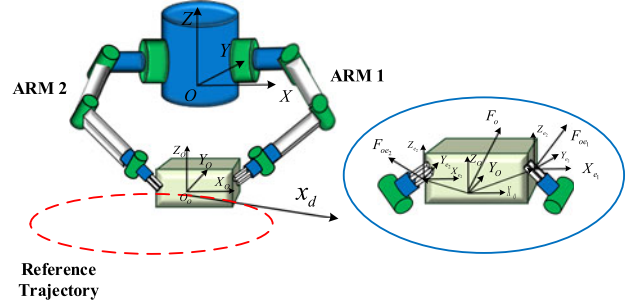


Fig. 1. Overview of the dual-arm robot manipulated a common object.

## II. PROBLEM FORMULATION AND MODELING PROCEDURE

### A. Problem Formulation

Consider a bimanual robot grasping a common object; our objective is to design a robot controller such that the manipulated object could track a desired trajectory  $x_d$  specified in the task space, as shown in Fig. 1, while simultaneously guarantee the following.

- 1) The tracking errors fall into the predefined bounds to achieve tailored tracking performances.
- 2) All the signals in the closed-loop bimanual robot system remain GUUB.
- 3) The internal forces between the end-effectors and the object converge to a small neighborhood of specified values.

### B. Modeling of the Bimanual Robot

The position and orientation of the manipulated object could be defined by a vector  $x \in \mathbb{R}^{N_0}$ , where  $N_0$  is the object's degree of freedom (DOF). Assume that both arms grasp the object rigidly so that there is no relative motion in between the object and the end-effectors. Then, based on the forward kinematics of robot manipulator, the relations between task space and robot joint space can be calculated in the following manner:

$$x = p_i(q_i), \quad \dot{x} = \dot{p}_i(q_i) = J_i(q_i)\dot{q}_i \quad (1)$$

where  $q_i \in \mathbb{R}^{N_i}$  and  $\dot{q}_i \in \mathbb{R}^{N_i}$  are vectors of joint variable and joint velocity of the  $i$ th robotic arm, respectively, and  $N_i$  is the DOF of the  $i$ th robotic arm.  $p_i$  is a continuous function, and  $J_i(q_i)$  is the Jacobian matrix. The following assumptions are considered to facilitate the modeling procedure of the bimanual robot system.

**Assumption 1:** The dynamics of the robot manipulators are uncertain, while the kinematics is accurately available. The robotic arms are operating away from any singular configurations during the motion.

**Assumption 2:** The rigid object would not be deformed by the exerted forces.

Then, the dynamics of each robot arm are described in the following Lagrangian form:

$$M_i(q_i)\ddot{q}_i + C_i(q_i, \dot{q}_i)\dot{q}_i + G_i(q_i) = \tau_i + J_{e_i}^T(q_i)F_{e_i} \quad (2)$$

where  $M_i(q_i) \in \mathbb{R}^{N_i \times N_i}$ ,  $C_i(q_i, \dot{q}_i) \in \mathbb{R}^{N_i \times N_i}$ , and  $G_i(q_i) \in \mathbb{R}^{N_i}$  are the inertial matrix, Coriolis and centrifugal matrix, and

gravity vector, respectively.  $J_{e_i}^T(q_i)$  represents the robotic arm's Jacobian matrix, while  $\tau_i \in \mathbb{R}^{N_i}$  is the joint torque, and  $F_{e_i} \in \mathbb{R}^{N_0}$  is the force vector exerted at end-effector. The dynamics of the object's motion can be described as

$$M_o(x)\ddot{x} + C_o(x, \dot{x})\dot{x} + G_o(x) = F_o \quad (3)$$

where  $M_o(x)$ ,  $C_o(x, \dot{x})$ , and  $G_o(x)$  denote the inertial, Coriolis and centrifugal matrix, and the gravitational vector of manipulated object, respectively, while  $F_o \in \mathbb{R}^{N_0}$  is the resulting force given as follows:

$$F_o = -F_{oe1} - F_{oe2}, \quad F_{oei} = f_i + f_{oi} \quad (4)$$

where  $F_{oei}$  is the interaction force applied on the end-effector of the  $i$ th robotic arm.  $F_{oei}$  are decomposed into an external force  $f_{oi}$  and an internal force  $f_i$ , where the external forces  $f_{oi}$  derive the motion of the object, and the internal forces  $f_i$  cancel with each other and satisfy the constraint  $f_1 + f_2 = 0_{[n]}$ . Combination of (3) and (4) yields

$$f_i = F_{oei} - D_i(t)(f_{o1} + f_{o2}) \quad (5)$$

where  $D_i(t) \in \mathbb{R}^{N_0 \times N_0}$  is the object load distribution matrix satisfying  $D_1(t) + D_2(t) = I_{N_0}$ , where  $I_{N_0} \in \mathbb{R}^{N_0 \times N_0}$  is an identity matrix.

Combination of (2)–(5) and the kinematic (1) yields a compact form below:

$$\tau_i = \mathcal{M}_i(q_i)\ddot{q}_i + \mathcal{C}_i(q_i, \dot{q}_i)\dot{q}_i + \mathcal{G}_i(q_i) - J_i^T(q_i)f_i \quad (6)$$

where  $\mathcal{M}_i = M_i + D_i\mathcal{M}_o$ ,  $\mathcal{M}_o = J_i^T M_o J_i$ ,  $\mathcal{C}_i = C_i + D_i(M_D + C_o)$ ,  $\mathcal{C}_o = J_i^T C_o J_i$ ,  $M_D = J_i^T \mathcal{M}_o J_i$ ,  $\mathcal{G}_i = G_i + D_i\mathcal{G}_o$ ,  $\mathcal{G}_o = J_i^T G_o$ . To be self-contained, the fundamental properties of robot manipulator dynamics, which will be used later for control design and analysis, are described below.

**Property 1 [10]:** The skew-symmetric matrix  $2\mathcal{C}_i(q_i, \dot{q}_i) - [\dot{\mathcal{M}}_i(q_i) - \dot{D}_i(t)\mathcal{M}_o(q_i, \dot{q}_i)]$  satisfies that

$$\partial^T \left\{ (2\mathcal{C}_i(q_i, \dot{q}_i) - \dot{\mathcal{M}}_i(q_i)) - \dot{D}_i(t)\mathcal{M}_o(q_i, \dot{q}_i) \right\} \partial = 0, \quad \forall \partial.$$

**Property 2 [10]:** The matrix  $\dot{D}_i(t)\mathcal{M}_o(q_i)$  is bounded and uniformly continuous and satisfies the following inequality:

$$\|\dot{D}_i(t)\mathcal{M}_o(q_i)\| \leq 2\varrho, \quad \forall t \geq 0 \quad (7)$$

where  $\varrho$  is a positive constant.

### III. CONTROL DESIGN

Before proceeding to the control design, let us introduce the following tracking error signals:

$$e = x - x_d, \quad z_i = \dot{q}_i - \alpha_i \quad i = 1, 2 \quad (8)$$

where  $e = [e_1, e_2, \dots, e_{N_0}] \in \mathbb{R}^{N_0}$  stands for the position tracking error of the manipulated object,  $z_i = [z_{i1}, z_{i2}, \dots, z_{iN_i}] \in \mathbb{R}^{N_i}$  stand for the velocity tracking error of each robotic arm in joint space,  $\alpha_i$  is a virtual controller to be specified in (19), and  $x_d$  is the reference trajectory of the manipulated object. Our control strategy is illustrated in Fig. 2.

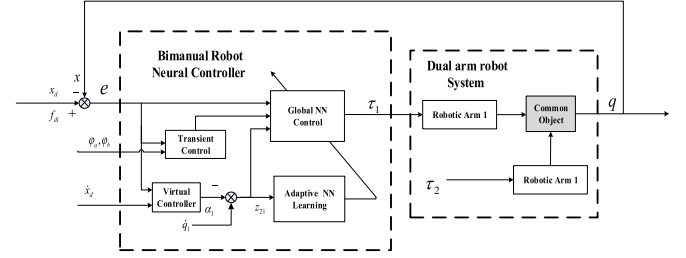


Fig. 2. Framework for the bimanual robot controller.

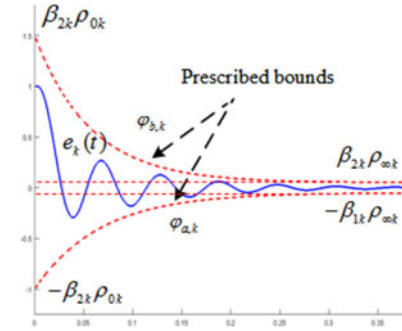


Fig. 3. Relationship between the tracking error  $e_k(t)$  and the performance function.

#### A. Specification on Requirement for Tracking Performance

To specify tracking performance, especially transient performance (e.g., overshoot, undershoot, and coverage rate), we construct a series of smoothly decreasing functions  $\phi(t) = [\phi_1, \phi_2, \dots, \phi_{N_0}]$  to shape the motion of the object as

$$\phi_k(t) = (\rho_{0k} - \rho_{\infty k})e^{-a_k t} + \rho_{\infty k} \quad (9)$$

where  $\rho_{0k}$ ,  $\rho_{\infty k}$ , and  $a_k$  ( $k = 1, 2, \dots, N_0$ ) are properly chosen positive constants. Let us define  $\varphi_{a,k}(t) = -\beta_{1k}\phi_k(t)$  and  $\varphi_{b,k}(t) = \beta_{2k}\phi_k(t)$ , with positive constants  $\beta_{1k}$  and  $\beta_{2k}$  to be specified by the designer.

**Remark 1:** The functions  $\varphi_{a,k}(t)$  and  $\varphi_{b,k}(t)$  specify the tracking transient response, i.e., the exponential term  $a_k$  regulates the required convergence rate of tracking errors,  $\beta_{1k}\rho_{0k}$  and  $-\beta_{2k}\rho_{0k}$  define the maximum overshoot and undershoot, and  $-\beta_{1k}\rho_{\infty k}$  and  $\beta_{2k}\rho_{\infty k}$  regulate the bounds of the steady errors, as shown in Fig. 3. This implies that we are able to regulate both transient and steady-state performance by properly choosing parameters  $\beta_{1k}$ ,  $\beta_{2k}$ ,  $\rho_{0k}$ ,  $\rho_{\infty k}$ , and  $a_k$ .

The following coordinate transformation of tracking errors will be used in the later design:

$$\xi_a = \left[ \frac{e_1}{\varphi_{a,1}}, \dots, \frac{e_{N_0}}{\varphi_{a,N_0}} \right]^T, \quad \xi_b = \left[ \frac{e_1}{\varphi_{b,1}}, \dots, \frac{e_{N_0}}{\varphi_{b,N_0}} \right]^T \quad (10)$$

$$\xi_k = h_k(e_k)\xi_{b,k} + (1 - h_k(e_k))\xi_{a,k}$$

where  $\xi_{a,k}$  and  $\xi_{b,k}$  are the  $k$ th element of the vectors  $\xi_a$  and  $\xi_b$ , respectively, and  $h_k(e_k)$  is defined as

$$h_k(e_k) = \begin{cases} 1, & e_k \geq 0 \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

## B. Controller Design Using BLF and Backstepping

Inspired by the work [40], an asymmetric time-varying barrier function is constructed for the  $i$ th robotic arm as

$$V_{i1} = \sum_{k=1}^{N_0} \left( \frac{h_k}{2} \ln \frac{1}{1 - \xi_{b,k}^2} + \frac{1 - h_k}{2} \ln \frac{1}{1 - \xi_{a,k}^2} \right). \quad (12)$$

The differentiation of (12) with respect to time gives us

$$\dot{V}_{i1} = \sum_{k=1}^{N_0} \left( \frac{h_k}{1 - \xi_{b,k}^2} \xi_{b,k} \dot{\xi}_{b,k} + \frac{1 - h_k}{1 - \xi_{a,k}^2} \xi_{a,k} \dot{\xi}_{a,k} \right). \quad (13)$$

According to definitions of  $\xi_{a,k}$  and  $\xi_{b,k}$ , and substituting (8) into (13), we have

$$\begin{aligned} \dot{V}_{i1} = & \sum_{k=1}^{N_0} \left( \frac{\xi_k^2}{(1 - \xi_k^2) e_k} \dot{e}_k \right) \\ & + \sum_{k=1}^{N_0} \left( \frac{(1 - h_k) \xi_{a,k}^2}{(1 - \xi_{a,k}^2)} \frac{\dot{\varphi}_{a,k}}{\varphi_{a,k}} + \frac{h_k \xi_{b,k}^2}{(1 - \xi_{b,k}^2)} \frac{\dot{\varphi}_{b,k}}{\varphi_{b,k}} \right). \end{aligned} \quad (14)$$

Then, by defining a transient control vector

$$P = \left[ \frac{\xi_1^2}{(1 - \xi_1^2) e_1}, \frac{\xi_2^2}{(1 - \xi_2^2) e_2}, \dots, \frac{\xi_{N_0}^2}{(1 - \xi_{N_0}^2) e_{N_0}} \right]^T \quad (15)$$

and substituting it into (14), we rewrite  $V_{i1}$  as follows:

$$\dot{V}_{i1} = P^T \dot{e} + \sum_{k=1}^{N_0} \left( \frac{(1 - h_k) \xi_{a,k}^2}{(1 - \xi_{a,k}^2)} \frac{\dot{\varphi}_{a,k}}{\varphi_{a,k}} + \frac{h_k \xi_{b,k}^2}{(1 - \xi_{b,k}^2)} \frac{\dot{\varphi}_{b,k}}{\varphi_{b,k}} \right). \quad (16)$$

Note that the relation between  $\dot{x}$  and  $\dot{q}_i$  as specified in (1) always hold. According to the definitions of  $e$  and  $z_i$  in (8), we have

$$\dot{e} = J_i(q)(z_i + \alpha_i) - \dot{x}_d, \quad i = 1, 2. \quad (17)$$

Substituting (17) into (16) yields

$$\begin{aligned} \dot{V}_{i1} = & P^T (J_i(q)(z_i + \alpha_i) - \dot{x}_d) \\ & + \sum_{k=1}^{N_0} \left( \frac{(1 - h_k) \xi_{a,k}^2}{(1 - \xi_{a,k}^2)} \frac{\dot{\varphi}_{a,k}}{\varphi_{a,k}} + \frac{h_k \xi_{b,k}^2}{(1 - \xi_{b,k}^2)} \frac{\dot{\varphi}_{b,k}}{\varphi_{b,k}} \right). \end{aligned} \quad (18)$$

Then, let us design a virtual controller  $\alpha_i$  as

$$\alpha_i = J_i^+(q) (\dot{x}_d - K_1 e - \sigma(t)e) \quad (19)$$

where  $J_i^+(q_i)$  is the Moore–Penrose inverse of  $J_i(q_i)$ ,  $K_1 = \text{diag}\{k_{11}, k_{12}, \dots, k_{1N_0}\}$ , with  $k_{1k}$  being positive constants. And  $\sigma(t) = \text{diag}\{\sigma_1(t), \sigma_2(t), \dots, \sigma_{N_0}(t)\}$  with  $\sigma_k(t) = \sqrt{(\frac{\dot{\varphi}_{a,k}}{\varphi_{a,k}})^2 + (\frac{\dot{\varphi}_{b,k}}{\varphi_{b,k}})^2} + k_a$ , where  $k_a$  selected as a positive parameter that ensures the boundedness of  $\dot{\alpha}_i$  when  $\dot{\varphi}_{a,k}(t)$  and

$\dot{\varphi}_{b,k}(t)$  are zero. Substituting (19) into (18) yields

$$\begin{aligned} \dot{V}_{i1} = & P^T J_i(q) z_i - P^T (K_1 e + \sigma(t)e) \\ & + \sum_{k=1}^{N_0} \left( \frac{h_k \xi_{b,k}^2}{(1 - \xi_{b,k}^2)} \left( \frac{\dot{\varphi}_{b,k}}{\varphi_{b,k}} \right) + \frac{(1 - h_k) \xi_{a,k}^2}{(1 - \xi_{a,k}^2)} \left( \frac{\dot{\varphi}_{a,k}}{\varphi_{a,k}} \right) \right). \end{aligned} \quad (20)$$

Note that the following inequality holds:

$$\sigma_k(t) - h_k \frac{\dot{\varphi}_{a,k}}{\varphi_{a,k}} - (1 - h_k) \frac{\dot{\varphi}_{b,k}}{\varphi_{b,k}} \geq 0. \quad (21)$$

Using the definition of  $P$  in (15) and in terms of (21), (20) can be rewritten as

$$\dot{V}_{i1} \leq - \sum_{k=1}^{N_0} k_{1k} \frac{\xi_k^2}{(1 - \xi_k^2)} + P^T J_i(q) z_i. \quad (22)$$

## C. Global Adaptive NN (GANN) Control

### 1) Radial Basis Function Neural Network (RBFNN)

[42]: In this paper, the following RBFNNs are used to approximate a continuous vector function  $F(Z) = [f_1(Z), f_2(Z), \dots, f_n(Z)]^T \in \mathbb{R}^n$ ,

$$\hat{F}(Z) = \hat{W}^T S(Z) \quad (23)$$

where  $\hat{F}(Z) \in \mathbb{R}^n$  is the estimate of  $F(Z)$ ,  $Z \in \Omega_Z \subset \mathbb{R}^q$  is NN inputs vector, and  $q$  denotes the demonstration of the input;  $\hat{W} = [\hat{W}_1, \hat{W}_2, \dots, \hat{W}_n] \in \mathbb{R}^{n \times l}$  is the estimation of NN optimal weight matrix  $W^*$ , and  $l$  is the number of NN nodes.  $S(Z) = [s_1(Z), s_2(Z), \dots, s_l(Z)]^T \in \mathbb{R}^l$  is the regressor vector, with  $s_i(\cdot)$  being a radial basis function. In general, the most commonly used Gaussian radial basis functions are employed as follows:

$$s_i(\|Z - \mu_i\|) = \exp \left[ \frac{-(Z - \mu_i)^T (Z - \mu_i)}{\vartheta_i^2} \right] \quad (24)$$

where  $\mu_i$  ( $i = 1, \dots, l$ ) are distinct points in state space,  $\mu_i = [\mu_{i1}, \mu_{i2}, \dots, \mu_{iq}]^T$  is the center of the neural and  $\vartheta_i$  is the Gaussian function's width. It has been established that, with sufficiently large node number, an arbitrary continuous function  $F(Z)$  can be approximated by the RBFNN (23) over a compact set  $\Omega_Z$  as

$$F(Z) = W^{*T} S(Z) + \varepsilon(Z) \quad \forall Z \in \Omega_Z \quad (25)$$

where  $W^*$  is an ideal constant weight vector, and  $\varepsilon(Z) \in \mathbb{R}^n$  is the approximation error. There exist ideal weight vector  $W^*$  such that  $|\varepsilon(Z)| < \varepsilon^*$  with constant  $\varepsilon^* > 0$  for all  $Z \in \Omega_Z$ .

**2) Global NN Control Design:** Let us define a positive Lyapunov function as

$$V_{i2} = V_{i1} + \frac{1}{2} z_i^T \mathcal{M}_i z_i. \quad (26)$$

Substituting (6) and (8) into its derivative, and considering Properties 1 and 2, we can derive from (26) that

$$\begin{aligned} \dot{V}_{i2} \leq & \dot{V}_{i1} + \varrho_i z_i^T z_i \\ & + z_i^T (\tau_i - \mathcal{M}_i \dot{\alpha}_i - \mathcal{C}_i \alpha_i - \mathcal{G}_i + J_i^T(q_i) f_i) \end{aligned} \quad (27)$$

where  $\mathcal{M}_i$ ,  $\mathcal{G}_i$ , and  $\mathcal{C}_i$  are abbreviations of  $\mathcal{M}_i(q)$ ,  $\mathcal{G}_i(q)$ , and  $\mathcal{C}_i(q, \dot{q})$ , respectively, and  $\varrho_i$  is a positive constant specified in (7).

Considering the dynamics of robot in (6), we reformulate it by using a function vector  $F_i(Z_i) \in \mathbb{R}^{N_i}$  as

$$F_i(Z_i) = -(\mathcal{M}\dot{\alpha}_i + \mathcal{C}_i\alpha_i + \mathcal{G}_i) \quad (28)$$

where  $F_i(Z_i) = [f_{i,1}(Z_i), f_{i,2}(Z_i), \dots, f_{i,N_i}(Z_i)]^T$ ,  $Z_i = [q_i^T, \dot{q}_i^T, \alpha_i^T, \dot{\alpha}_i^T]^T \in \mathbb{R}^{\nu_i}$ , with  $\nu_i = 4N_i$ . It should be noted that, for the functions  $f_{i,j}(Z_i) \in \mathbb{R}$ ,  $j = 1, 2, \dots, N_i$ , there exist known bounded nonnegative smooth functions  $f_{i,j}^U(Z_i)$  such that  $|f_{i,j}(Z_i)| \leq f_{i,j}^U(Z_i)$ ,  $\forall Z \in \mathbb{R}^{\nu_i}$ .

Applying RBFNN described in Section III-C, we see that over a compact set  $\Omega_{i1}$

$$\hat{F}_i(Z_i) = \hat{W}_i^T S_i(Z_i) + \varepsilon_i \quad (29)$$

where  $\hat{W}_i = [\hat{W}_{i,1}, \hat{W}_{i,2}, \dots, \hat{W}_{i,N_i}]^T \in \mathbb{R}^{l_i \times N_i}$  is the estimation of optimal neural weight matrix  $W_i^*$ , and  $\hat{W}_{i,j} = [\hat{\omega}_{i,j,1}, \hat{\omega}_{i,j,2}, \dots, \hat{\omega}_{i,j,l_i}] \in \mathbb{R}^{l_i}$ ,  $S_i(Z_i) \in \mathbb{R}^{l_i}$  is the basis vector function, with  $l_i$  being the NN node number, and  $\varepsilon_i$  is the NN construction error satisfying  $|\varepsilon_i| < \bar{\varepsilon}_i$ .

Prior to proceeding to the control design, let us introduce a set of smooth switching functions  $Q_i(Z_i) \in \mathbb{R}^{N_i \times N_i}$  as

$$Q_i(Z_i) = \text{diag}(M_{i1}(Z_i), M_{i2}(Z_i), \dots, M_{iN_i}(Z_i)) \quad (30)$$

where  $M_{ij}(Z_i) = \prod_{c=1}^{\nu_i} m(z_{ic})$ , and  $m(z_{ic})$  is designed as

$$m(z_{ic}) = \begin{cases} 1, & |z_{ic}| < d_{1,ic} \\ \frac{d_{2,ic}^2 - z_{ic}^2}{d_{2,ic}^2 - d_{1,ic}^2} e^{\left( \frac{z_{ic}^2 - d_{1,ic}^2}{\omega_i (d_{2,ic}^2 - d_{1,ic}^2)} \right)^2}, & \text{otherwise} \\ 0, & |z_{ic}| > d_{2,ic} \end{cases} \quad (31)$$

where  $d_{1,ic}$  and  $d_{2,ic}$  are positive constants satisfying  $0 < d_{1,ic} < d_{2,ic}$ , and  $\omega_i$  are positive constants with  $\omega_i \geq 1$ .

**Remark 2:** The switching function  $m(\cdot)$  is scaled to  $m(\cdot) = 1$  in the compact set  $\Omega_1$  and  $m(\cdot) = 0$  outside the domain  $\Omega_2$ , as shown in Fig. 4. Therefore, the adaptive NN control can be thoroughly disabled when the neural active region is no longer remain.

Then, the adaptive global NN robot control law is designed as

$$\tau_i = -K_{2i}z_i - J_i^T(q_i)P - J_i^T(q_i)f_{di} - Q_i(Z_i)\Phi_i^a - (1 - Q_i(Z_i))\Phi_i^b \quad (32)$$

where  $K_{2i} = \text{diag}\{k_{2,i1}, k_{2,i2}, \dots, k_{2,iN_i}\}$  is an designed positive definite diagonal matrix,  $f_{di}$  is the desired internal force, and  $P$  is the transient controller specified in (15).  $\Phi_i^a$  and  $\Phi_i^b$  are designed as

$$\Phi_i^a = \hat{F}_i(Z_i), \quad \Phi_i^b = F_i^U(Z_i)\Gamma_i \left( \frac{F_i^U(Z_i)z_i}{\varpi} \right) \quad (33)$$

where  $\hat{F}_i$  is the estimate of  $F_i$ , and  $F_i^U = \text{diag}\{f_{i,1}^U(Z_i), f_{i,2}^U(Z_i), \dots, f_{i,N_i}^U(Z_i)\}$ ,  $\Gamma_i \left( \frac{F_i^U(Z_i)z_i}{\varpi} \right) = [\tanh(\frac{F_i(Z_i)z_{i1}}{\varpi}), \tanh(\frac{F_i(Z_i)z_{i2}}{\varpi}), \dots, \tanh(\frac{F_i(Z_i)z_{iN_i}}{\varpi})]^T$ , with  $\varpi$  being a positive parameter.

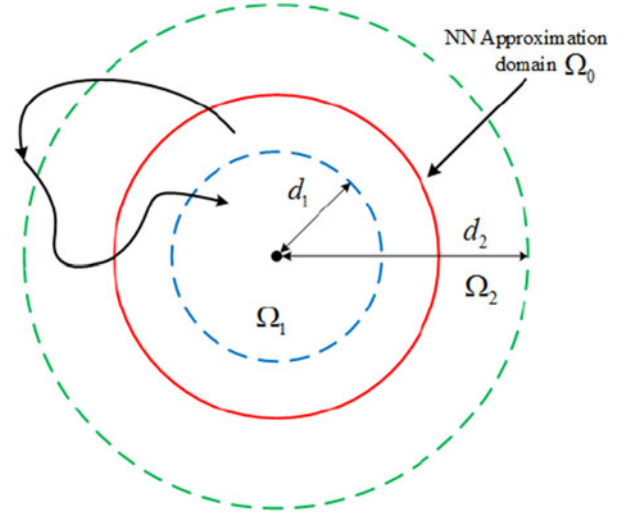


Fig. 4. Global tracking performance.

The NN weight adaptive law is designed as

$$\dot{\hat{W}}_i = \Theta_i(Q_i(Z_i)S_i(Z_i)z_i - \gamma_i\hat{W}_i) \quad (34)$$

where  $\Theta_i$  is a positive-definite matrix, and  $\gamma_i$  is a positive constant.

**Remark 3:** The controller proposed in (32) consists of an adaptive NN controller  $\Phi_i^a$  and an extra robust controller  $\Phi_i^b$ . When the tracking runs in the NN active domain  $\Omega_1$ , the term  $\Phi_i^a$  plays a decisive role; once the NN runs out of the  $\Omega_2$ , the extra robust term  $\Phi_i^b$  will pull the state back. If the NN runs in the domain between the  $\Omega_2$  and  $\Omega_1$ , both terms work and will pull the state back to the compact set  $\Omega_1$ .

Consider the following Lyapunov function:

$$V_i = V_{i2} + \frac{1}{2} \sum_{j=1}^{N_i} \tilde{W}_{i,j}^T \Theta_i^{-1} \tilde{W}_{i,j} \quad (35)$$

where  $(\tilde{*}) = (\hat{*}) - (*)$ . Taking the derivative of (35) along time, and considering the control law (32) and the adaptive law (34), yields

$$\begin{aligned} \dot{V}_i &= \dot{V}_{i1} + z_i^T \left( -K_{2i}z_i + \varrho_i z_i - J_i^T(q_i)P + J_i^T(q_i)\tilde{f}_i \right) \\ &\quad + z_i^T \left( -Q_i(Z_i)\Phi_i^a - (I - Q_i(Z_i))\Phi_i^b + F_i(Z_i) \right) \\ &\quad + \sum_{j=1}^{N_i} \tilde{W}_{i,j}^T \Theta_i^{-1} \dot{\tilde{W}}_{i,j} \end{aligned} \quad (36)$$

where  $\tilde{f}_i = f_i - f_{di}$ . Substituting (29), (33), and (34) into (36), we have

$$\begin{aligned} \dot{V}_i &\leq \dot{V}_{i1} + z_i^T (-K_{2i}z_i - J_i^T(q_i)P + J_i^T(q_i)\tilde{f}_i + \varrho_i z_i) \\ &\quad + \sum_{j=1}^{N_i} \left( z_{ij} f_{i,j}^U - z_{ij} f_{i,j}^U \tanh \left( \frac{z_{ij} f_{i,j}^U}{\varpi} \right) \right) \\ &\quad + \sum_{j=1}^{N_i} \left( -\gamma_i \tilde{W}_{i,j}^T (W_{ij}^* + \tilde{W}_{i,j}) + z_{ij} \varepsilon_{ij} \right). \end{aligned} \quad (37)$$

Note that the following inequalities hold in terms of Young's inequality:

$$\begin{aligned} -\tilde{W}_{i,j}^T(W_{ij}^* + \tilde{W}_{i,j}) &\leq -\frac{1}{2}\|\tilde{W}_{i,j}\|^2 + \frac{1}{2}\|W_{i,j}^*\|^2 \\ z_{ij}\varepsilon_{ij} &\leq \frac{1}{2}z_{ij}^2 + \frac{1}{2}\varepsilon_{ij}^2. \end{aligned} \quad (38)$$

And the following inequality holds for any  $\varpi > 0$  and  $z \in \mathbb{R}$ :

$$0 \leq |z| - z \tanh\left(\frac{z}{\varpi}\right) \leq \kappa\varpi \quad (39)$$

where  $\kappa$  is a constant satisfying  $\kappa = e^{-(\kappa+1)}$ , i.e.,  $\kappa = 0.2785$ .

Substituting (22), (38), and (39) into (37), we have

$$\begin{aligned} \dot{V}_i &\leq -\sum_{k=1}^{N_0} k_{1j} \frac{\xi_k^2}{(1-\xi_k^2)} + \left(\dot{e} + (K_1 + \sigma)e\right)^T \tilde{f}_i \\ &\quad + \sum_{j=1}^{N_i} \left(-\left(k_{2,ij} - \varrho_i - \frac{1}{2}\right)z_{ij}^2 - \frac{1}{2}\gamma_i \|\tilde{W}_{i,j}\|^2\right) \\ &\quad + \sum_{j=1}^{N_i} \left(\frac{1}{2}\gamma_i \|W_{i,j}^*\|^2 + \frac{1}{2}\varepsilon_{ij}^2 + \kappa\varpi\right). \end{aligned} \quad (40)$$

Then, taking the Lyapunov function  $V = V_1 + V_2$  and considering the property of internal forces, we have

$$\begin{aligned} \dot{V} &= \dot{V}_1 + \dot{V}_2 \leq \sum_{k=1}^{N_0} \left(-2k_{1k} \ln \frac{1}{(1-\xi_k^2)}\right) \\ &\quad + \sum_{i=1}^2 \sum_{j=1}^{N_i} \left(-k_{c,ij} z_{ij}^2 - \frac{1}{2}\gamma_i \|\tilde{W}_{i,j}\|^2\right) \\ &\quad + \sum_{i=1}^2 \sum_{j=1}^{N_i} \left(\frac{1}{2}\gamma_i \|W_{i,j}^*\|^2 + \frac{1}{2}\varepsilon_{ij}^2 + \kappa\varpi\right) \end{aligned} \quad (41)$$

where  $k_{c,ij} = k_{2,ij} - \varrho_i - \frac{1}{2}$ , and the fact  $\frac{\xi_k^2}{(1-\xi_k^2)} \geq \ln \frac{1}{(1-\xi_k^2)}$ ,  $\forall |\xi_k| < 1$  has been used.

### D. Stability Analysis

**Theorem 1:** Consider the bimanual robot system in (6), together with the virtual controllers  $\alpha_i$  in (19), the control law (32), the adaptation law in (34), and the performance functions in (9). Given initial conditions  $e_k$  satisfy  $\varphi_{a,k}(0) < e_k(0) < \varphi_{b,k}(0)$ , the proposed adaptive control scheme can guarantee the following.

- 1) The tracking error  $e$  are bounded by the predefined function  $\varphi_{aj}$ .
- 2) All the tracking signals in the closed-loop system are uniformly ultimately bounded.
- 3) The tracking error  $e$  converges to a small neighborhood of zero.

*Proof.* From (12), (26), and (35), we have

$$\begin{aligned} V &= \sum_{k=1}^{N_0} \left(\ln \frac{1}{1-\xi_k^2}\right) + \frac{1}{2} \sum_{i=1}^2 z_i^T \mathcal{M}_i(z_i) z_i \\ &\quad + \frac{1}{2} \sum_{i=1}^2 \sum_{j=1}^{N_i} \tilde{W}_{i,j}^T \Theta_i^{-1} \tilde{W}_{i,j}. \end{aligned} \quad (42)$$

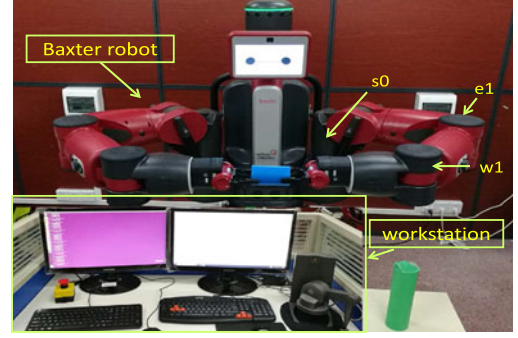


Fig. 5. Illustration of the setup of the experiment. [photo taken at South China University of Technology].

According to (42), inequality (41) can be represented as

$$\dot{V}(t) = -\eta V(t) + \mu \quad (43)$$

where  $\eta = \min\{2\lambda_{\min}(K_1), \frac{2\lambda_{\min}(K_{c_i})}{\lambda_{\max}(\mathcal{M}_i)}, \frac{\gamma_i}{\lambda_{\max}(\Theta_i^{-1})}, |i = 1, 2\}$ ,  $\mu = \sum_{i=1}^2 \sum_{j=1}^{N_i} (\frac{1}{2}\varepsilon_{ij}^2 + \frac{1}{2}\|W_{i,j}^*\|^2 + \kappa\varpi)$ , and  $K_{c_i} = \text{diag}\{k_{c,i1}, k_{c,i2}, \dots, k_{c,iN_i}\}$ .

Multiplying both sides by  $e^{\eta t}$  in (43), and applying the integration over  $[0, t]$ , we have

$$V(t) \leq (V(0) - \mu/\eta)Ee^{-\eta t} + \mu/\eta \leq V(0) + \mu/\eta. \quad (44)$$

From the above inequality, and in terms of (12), (26), and (35), as well as the initial condition of  $\xi_k(0)$ , we can conclude that the terms  $\ln(1/(1-\xi_k^2))$ ,  $z_i$  as well as the NN weight estimation errors  $\tilde{W}_{i,j}$  are bounded. Thus, we can conclude that  $\varphi_a < e < \varphi_b$ , which implies the transient performance is guaranteed. And since  $\varphi_a$  and  $\varphi_b$  are bounded function,  $e$  must be bounded. From (8), we can obtain that  $x$  is also bounded. Therefore  $J_i$  is bounded. From the definition of  $\alpha_i$ , we can know that  $\alpha_i$  is also bounded. In terms of the boundedness of  $z_i$  and  $\alpha_i$  and according to  $\dot{q}_i = z_i + \alpha_i$ ,  $\dot{q}_i$  is also bounded. Hence, all the signals in the closed-loop dual-arm robot system are bounded. This completes the proof. ■

**Remark 4:** The designed matrices  $K_1$  and  $K_{c_i}$  in the controller can be chosen simply as positive-definite diagonal matrices. The gains in the NN adaptive law  $\Theta_i$  and  $\gamma_i$  should be positive. And in terms of (44), if the gains  $K_1$ ,  $K_{2_i}$ , and  $\gamma_i$  are chosen to be relatively small, while  $\Theta_i$  chosen relatively large, then the amplitude of tracking error could be made smaller.

**Theorem 2:** The proposed global adaptive NN controller (32) also guarantees that the error of the internal force  $\tilde{f}_i$  converges to a small neighborhood of the origin.

*Proof.* See the Appendix.

## IV. EXPERIMENTAL STUDIES

The Baxter bimanual robot, as shown in Fig. 5, is used in the experiment. It is of two 7-DOF arms and advanced sensing technologies, including position, force, and torque sensors and control at every joint. The resolution for the joint sensors is 14 bits with  $360^\circ$  ( $0.022^\circ$  per tick resolution), while the maximum joint torques that can be applied to the joints are 50 N·m (the first four joints) and 15 N·m (the last three joints).

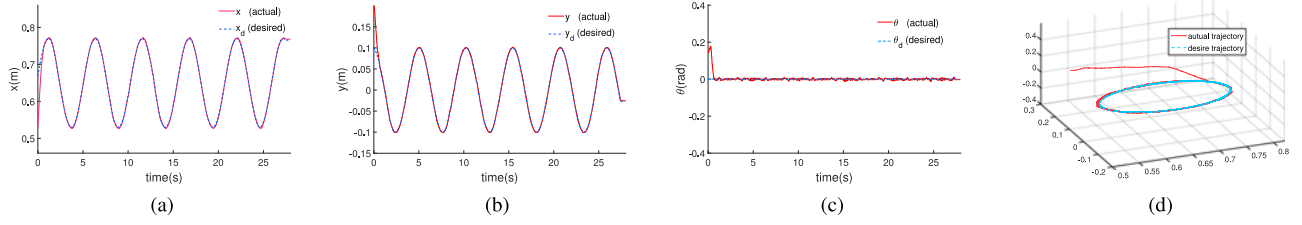


Fig. 6. Tracking performance of the manipulated object. (a)  $x$ . (b)  $y$ . (c)  $\theta$ . (d) Task space trajectory.

In the experiment, the Baxter robot is commanded to grasp an object by using its two robotics arms with grippers mounted on the end-effectors. For each robotic arm, we initialized the position of the joints to make the arm locating in a horizontal plane as shown in Fig. 5. For simplicity and without loss of generality, we use three parallel revolute joints ( $s_0, e_1, w_1$ ) of each arm to derive the motion in the experiment. The grasped object is a cylinder made of plastic, with 0.1 kg in weight, 0.1 m in length, and 0.06 m in diameter. The internal forces could be calculated by using torque sensors equipped with each joint together with gravity compensation model built in [43] and in terms of (5).

In order to well approximate the robot dynamics and considering both the accuracy and the computational efficiency, we divide the inputs of RBFNN into two groups, with one group containing  $[q_i^T, \dot{\alpha}_i^T]^T \in R^6$  and another  $[q_i^T, \dot{q}_i^T, \alpha_i^T]^T \in R^9$ , and employ three centers for each input dimension of the NNs, and ended up with totally  $l_1 = 20412$  NN nodes for each NN. The centers of the NNs nodes are evenly spaced between the upper and lower bounds of the motion range and speed limits of each joint in  $[-1.7, 1.7] \times [-1.05, 2.61] \times [-1.57, 2.09] \times [-1.5, 1.5] \times [-1.5, 1.5] \times [-1.5, 1.5] \cup [-1.7, 1.7] \times [-1.05, 2.61] \times [-1.57, 2.09] \times [-1.5, 1.5] \times [-1.5, 1.5] \times [-1.5, 1.5]$ . And the NNs weight matrix are initialized as  $\hat{W}_1(0) = \mathbf{0} \in \mathbb{R}^{3l_1 \times 3}$  and  $\hat{W}_2(0) = \mathbf{0} \in \mathbb{R}^{3l_2 \times 3}$ . And the gains of NN adaptive law are chosen as  $\Theta_1 = \text{diag}\{2\}$ ,  $\Theta_2 = \text{diag}\{2\}$ . The designed parameters  $K_1$  and  $K_{2i}$  of the controller are specified as  $K_1 = \text{diag}\{10, 9, 9\}$ ,  $K_{21} = K_{22} = \text{diag}\{9, 4.5, 1.2\}$ . And the parameters  $\varpi$  in the controller (32) are selected as  $\varpi = 0.1$ .

In the experiment, the object is required to trace the following trajectory specified in the Cartesian space:

$$\begin{pmatrix} x \\ y \\ \theta \end{pmatrix} = \begin{pmatrix} 0.65 + 0.1 \sin(2\pi/5t) \\ 0.12 \cos(2\pi/5t) \\ 0 \end{pmatrix}. \quad (45)$$

The initial configuration of the object is  $(0.55, 0.2, 0.2)$ , and the initial velocity is set to  $\dot{x}(0) = 0$ ,  $\dot{y}(0) = 0$ ,  $\dot{\theta}(0) = 0$ . The desired internal forces are chosen as  $f_{d1} = [0, 3, 0]$ ,  $f_{d2} = [0, -3, 0]$ . The parameters of the performance function (9) are designed with  $\rho_{01} = \rho_{02} = 0.2$ ,  $\rho_{03} = 0.4$ ,  $\rho_{\infty 1} = \rho_{\infty 2} = 0.012$ ,  $\rho_{\infty 3} = 0.025$ , and  $a_k = 2.5$ ,  $\beta_{1k} = \beta_{2k} = 1$ ,  $k = 1, 2, 3$ .

### A. Experimental Results

The experimental results are presented in Figs. 6–9. The tracking performance of the manipulated object in task space is shown

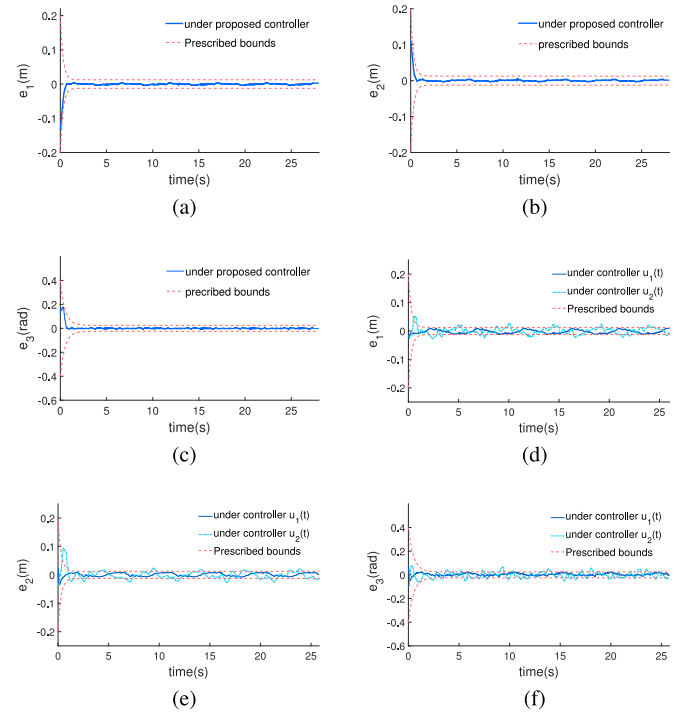
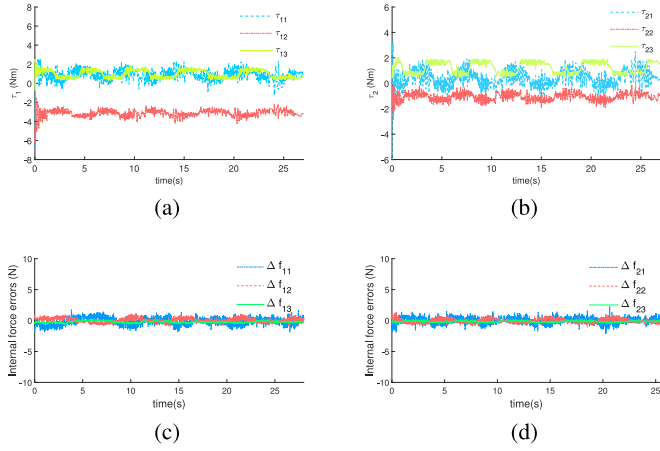
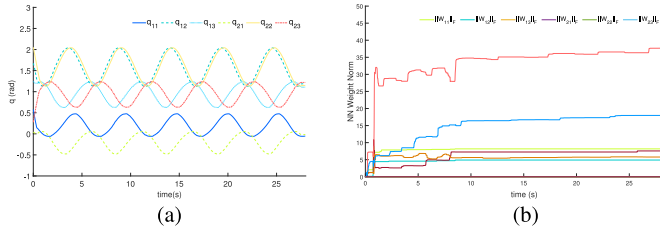


Fig. 7. Tracking error of the object manipulation. (a)–(c) Tracking errors with the proposed controller. (d)–(f) Tracking errors with the modified controllers  $u_1(t)$  and  $u_2(t)$ .

in Fig. 6(d), where the proposed controller is observed with a good performance when following a circular trajectory. The tracking trajectories with respect to  $x$ ,  $y$ , and  $\theta$  are depicted in Fig. 6(a)–(c). The tracking errors of the manipulated object are shown in Fig. 7(a)–(c). As shown in these figures, the grasped object follows the reference trajectories very well, and the tracking errors converge to a neighborhood around zero without violation of the prescribed transient bound (red-dashed line “-”). The trajectories of control inputs, internal force errors, joint positions, and NN weight norm are depicted as shown in Figs. 8 and 9. We can see from the figures that closed-loop signals are bounded and the internal force errors converge to a neighborhood of zero. In addition, comparative experimental results based on two modified controllers are shown in Fig. 7(d)–(f) ( $u_1(t)$  controller without NN adaptation;  $u_2(t)$  controller without both transient and NN control). As shown in these figures, without using the NN control and transient control, the tracking errors violated the prescribed transient bounds, while relatively larger steady-stage errors are observed without using the NN control. The experimental results illustrate that our proposed



**Fig. 8.** Control inputs and internal force errors of each arm of the Baxter robot. (a)  $\tau_1$ . (b)  $\tau_2$ . (c)  $\tilde{f}_1 = f_1 - f_{d1}$ . (d)  $\tilde{f}_2 = f_2 - f_{d2}$ .



**Fig. 9.** Trajectory of joint position and the convergence of NN weight. (a) Joint angles  $q_{ij}$ . (b) NN weight norm  $\|W_{ij}\|$  ( $i = 1, 2, j = 1, 2, 3$ ).

controller can successfully guarantee the tracking errors remaining in the predefined region and ensure the prescribed transient bounds to be never violated.

## V. CONCLUSION

In this paper, we designed an adaptive neural control for general dual-arm robot systems, with prescribed tracking performance and guaranteed global stability. By introduction of a set of boundary functions and then introducing them into the controller design, specified motion precision in both transient and steady states are achieved. The transient responses such as overshoot, settling time, and final tracking RBFNNs are employed to approximate the unknown dynamics of both the robot arms and the manipulated object. Semiglobal stability achieved by the conventional neural control has been extended to global stability by incorporation of a switching mechanism into the controller. The resulted neural control also ensures proper internal force applied on the object, as specified by the designer. Experiment studies have demonstrated the effectiveness of the proposed control scheme.

## APPENDIX

*Proof of Theorem 2:* Combining (6) and (32), we can obtain the error dynamics equation as

$$\begin{aligned} \mathcal{M}_i \dot{z}_i + C_i z_i + \mathcal{G}_i + \varepsilon_i + (I - Q_i)(-\Phi_i^a - \Phi_i^b) + K_{2i} z_i \\ = J^T(q_i) \tilde{f}_i - J_i^T(q_i) P. \end{aligned} \quad (46)$$

Then, multiplying  $J_i(q_i) \mathcal{M}_i^{-1}$  on both sides on the (46), we have

$$\begin{aligned} M_{c_i} \tilde{f}_i = J_i(q_i) \mathcal{M}_i^{-1} (I - Q_i) (-\Phi_i^a - \Phi_i^b) + M_{c_i} P \\ + J_i(q_i) \mathcal{M}_i^{-1} ((C_i + K_{2i}) z_i + \mathcal{G}_i + \varepsilon_i) + J_i(q_i) \dot{z}_i \end{aligned} \quad (47)$$

where  $M_{c_i} = J_i(q_i) \mathcal{M}_i^{-1} J_i^T(q_i)$ . Since  $J_i(q_i) z_i = \dot{e} + (-K_1 + \sigma)e$ , we can obtain that

$$\dot{J}_i(q_i) z_i + J_i(q_i) \dot{z}_i = \ddot{e} + \Lambda \dot{e} \quad (48)$$

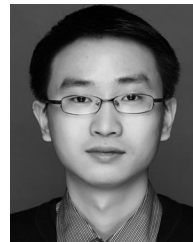
where  $\Lambda = -K_1 + \sigma$ . Substituting (48) into (46), we have  $M_{c_i} \tilde{f}_i = \ddot{e} + \Lambda \dot{e} + \chi_i$ , where  $\chi_i = J_i(q_i) \mathcal{M}_i^{-1} (I - Q_i) (-\Phi_i^a - \Phi_i^b) + M_{c_i} P + J_i(q_i) \mathcal{M}_i^{-1} ((C_i + K_{2i}) z_i + \mathcal{G}_i + \varepsilon_i) - \dot{J}_i(q_i) z_i$ .

Then, let us compute the term and consider the following equality by using the property of internal forces,  $U \tilde{f} = H$ , where  $\tilde{f} = [\tilde{f}_1^T, \tilde{f}_2^T]^T$ ,  $H = [\chi_1^T, \chi_2^T, 0]^T$ , and  $U = \begin{bmatrix} M_{c_1} & -M_{c_2} \\ I & I \end{bmatrix}$ . Since the terms  $M_{c_1}$  and  $M_{c_2}$  are positive definite, we can obtain that  $U$  is bounded and invertible. As analyzed in the proof of Theorem 1,  $q_i$ ,  $\dot{q}_i$ ,  $e$ , and  $z_i$  are all bounded, and we can deduce that  $\chi_i$  is also bounded; hence,  $H$  is bounded. Therefore, the vector of internal forces errors  $\tilde{f}$  are bounded. This completes the proof. ■

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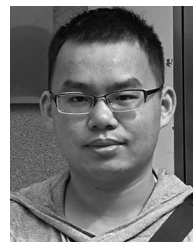
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